FINITE LOCALLY CONSTANT SHEAVES ARE EXACTLY THE SHEAVES REPRESENTED BY FINITE ETALE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both implications; the parenthetical about the construction of the descent datum refers to a routine transcription of an argument in the cited descent lemma, and every subsequent step of the descent is stated with its input.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Local structure of a finite etale morphism. Etale Morphisms, Lemma etale-lemma-finite-etale-etale-local: a finite etale morphism is locally a disjoint union of isomorphisms. Shows that a represented sheaf is finite locally constant.
- Effectivity of descent for schemes. Descent, Lemma descent-lemma-affine together with Lemmas descent-lemma-descending-property-finite and descent-lemma-descending-property-etale. Turns the constructed descent datum into a morphism of schemes and transfers finiteness and etaleness.
- The descent datum attached to a sheaf. Descent, Lemma descent-lemma-descent-data-sheaves, whose construction is copied. Produces the descent datum from the local representing objects.

Source reference. Original lemma 04HN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 9986–9986. Navigation pointer invoked at proof line 9986; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02W5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10011–10011. Navigation pointer invoked at proof

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line 10011; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0245: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10014–10014. Navigation pointer invoked at proof line 10014; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02LA: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10016–10016. Navigation pointer invoked at proof line 10016; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 02VN: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 10017–10017. Navigation pointer invoked at proof line 10017; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let X be a scheme and $\mathcal{F}$ a sheaf of sets on $X_{étale}$. Then the following are equivalent*

(1) *$\mathcal{F}$ is finite locally constant, and*
(2) *$\mathcal{F} = h_U$ for some finite étale morphism $U \to X$.*

Proof. A finite étale morphism is locally isomorphic to a disjoint union of isomorphisms, see Étale Morphisms, Lemma 04HN. Thus (2) implies (1). Conversely, if $\mathcal{F}$ is finite locally constant, then there exists an étale covering $\{X_i \to X\}$ such that $\mathcal{F}|_{X_i}$ is representable by $U_i \to X_i$ finite étale. Arguing exactly as in the proof of Descent, Lemma 02W5 we obtain a descent datum for schemes (U_i, φ_{ij}) relative to $\{X_i \to X\}$ (details omitted). This descent datum is effective for example by Descent, Lemma 0245 and the resulting morphism of schemes $U \to X$ is finite étale by Descent, Lemmas 02LA and 02VN. □